

COMP47670 Exam — Spring 2025/26

Guidelines

- Complete all three tasks. All tasks carry equal marks.
- All tasks should be completed in a single Jupyter Notebook. A template Jupyter notebook file has been provided on Brightspace.
- Use Markdown cells to briefly explain your work.
- A time limit of **2 hours** applies to this exam. You must submit your Jupyter notebook (IPYNB file) within this time.
- No collaboration is allowed. All submissions will be subject to plagiarism checking. Any evidence of plagiarism can result in a 0 grade.
- This is an open book exam (i.e., you can use module lecture/lab material and internet resources).
- Development environments (such as VSCode or Colab) **must not** be used for the exam. Use web-based Jupyter notebooks to complete the exam.

Use of Generative AI:

- Generative AI tools, code assistants, and large language models (such as ChatGPT, Copilot or Gemini) **must not** be used for any part of this exam.

Overview

This exam involves completing three tasks based on a synthetic dataset of daily air quality readings from monitoring stations in Dublin, where the goal is to predict whether a given day has poor air quality based on pollutant levels and weather conditions. The data is split across two CSV files which share a *date* column. Each row represents a single day, described by the following columns:

readings.csv

- *date*: date of the reading
- *pm10*: level of larger airborne particles in the air ($\mu\text{g}/\text{m}^3$)
- *pm25*: level of smaller airborne particles in the air ($\mu\text{g}/\text{m}^3$)
- *no2*: level of nitrogen dioxide in the air ($\mu\text{g}/\text{m}^3$)
- *so2*: level of sulphur dioxide in the air ($\mu\text{g}/\text{m}^3$)
- *co*: level of carbon monoxide in the air (mg/m^3)
- *poor_air_quality*: target label — 1 if the day had poor air quality, 0 if good

weather.csv

- *date*: date of the reading
- *temperature*: daily mean air temperature ($^{\circ}\text{C}$)
- *humidity*: daily mean relative humidity (%)
- *wind_speed*: daily mean wind speed (km/h)
- *rainfall*: total daily rainfall (mm)

The CSV files can be downloaded from the following links or from Brightspace:

- <http://mlg.ucd.ie/modules/python/readings.csv>
- <http://mlg.ucd.ie/modules/python/weather.csv>

Task 1: Preprocessing

- Load the two data files and merge them. Report the shape and the last 10 rows of the merged DataFrame.
- Convert *wind_speed* into a categorical feature by binning it into three groups using the following thresholds: Low (0–10 km/h), Moderate (10–20 km/h), and High (above 20 km/h). **Display a frequency table for the new categorical feature.**
- The *poor_air_quality* column is the target feature for classification in Task 3. Plot the class distribution based on this feature.

Task 2: Exploratory Visualisation

- Create a time series plot showing daily *pm10* and *pm25* feature values over the full date range. Plot both series together on a single figure.
- Using the same data, calculate a 14-day rolling average for both *pm10* and *pm25*. Plot both rolling averages together on a single figure.
- In Markdown, briefly discuss the trends in the time series plots from Tasks 2a and 2b.

Task 3: Classification

- Prepare the feature matrix and target vector. Split the data into training (70%) and test (30%) sets. **Report the class balance in both sets.**
- Train two different classifiers of your choice on the training set. Evaluate both on the test set using a **classification report** and confusion matrix.
- In Markdown, briefly discuss the evaluation results of both classifiers and **suggest how performance could be improved.**